

Jaeseung Heo

Graduate School of Artificial Intelligence
POSTECH (Pohang University of Science and Technology)

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EDUCATION

POSTECH

Integrated M.S./Ph.D. Program, Graduate School of Artificial Intelligence
Advisor: Prof. Dongwoo Kim

Pohang, South Korea
Mar, 2021 - Present
Expected Graduation: Feb, 2027

Sungkyunkwan University

B.S. in Computer Science and Engineering
B.S. in Systems Management Engineering (Double Major)

Seowon, South Korea
Mar, 2016 - Aug, 2020

RESEARCH INTERESTS

I study how training data shapes model behavior, viewing interpretability as a prerequisite for building safe AI systems. My work spans (i) influence-function-based training data attribution in graph neural networks, (ii) data-centric interventions such as augmentation and posterior label smoothing, and (iii) ongoing work on scalable analysis of training-example interactions in large language models. I am also interested in linking training data attribution to mechanistic interpretability by analyzing how data-level changes manifest in sparse autoencoder features and cross-layer transcoder features to explain specific model behaviors.

PUBLICATIONS

Conference Publications

- **Posterior Label Smoothing for Node Classification** AAAI '26
Jaeseung Heo, Moonjeong Park, Dongwoo Kim Oral Pres.
- **Influence Functions for Edge Edits in Non-Convex Graph Neural Networks** NeurIPS '25
Jaeseung Heo, Kyeongheung Yun, Seokwon Yoon, Moonjeong Park, Jungseul Ok, Dongwoo Kim
- **EPIC: Graph Augmentation with Edit Path Interpolation via Learnable Cost** IJCAI '24
Jaeseung Heo*, Seungbeom Lee*, Sungsoo Ahn, Dongwoo Kim
- **Mitigating Oversmoothing through Reverse Process of GNNs for Heterophilic Graphs** ICML '24
Moonjeong Park, Jaeseung Heo, Dongwoo Kim

*Equal contribution.

Preprints

- **Transductive Generalization and Its Application to Graph Node Classification** arXiv '26
MoonJeong Park, Seungbeom Lee, Kyungmin Kim, Jaeseung Heo, Seunghyuk Cho, Shouheng Li, Sangdon Park, Dongwoo Kim
- **The Oversmoothing Fallacy: A Misguided Narrative in GNN Research** arXiv '25
Moonjeong Park, Sunghyun Choi, Jaeseung Heo, Eunhyeok Park, Dongwoo Kim

REFERENCES

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